ITI – RedHat Admin 1 Lab 5

Mohammad Tarek – Data Management

1) **Compress a file by compress, gzip, zip commands and decompress it again. State the differences between compress and gzip commands.**

```
mohammadtarek@localhost:~
Mohammadgzip filetocompress
Mohammadzcat filetocompress.gz
BASH_BUILTINS(1)            General Commands Manual           BASH_BUILTINS(1)

NAME
       bash,  :,  .,  [, alias, bg, bind, break, builtin, caller, cd, command,
       compgen, complete, compopt, continue, declare, dirs, disown, echo,  en-
       able,  eval,  exec,  exit,  export, false, fc, fg, getopts, hash, help,
       history, jobs, kill, let, local, logout, mapfile, popd, printf,  pushd,
       pwd,  read, readonly, return, set, shift, shopt, source, suspend, test,
       times, trap, true, type, typeset, ulimit, umask, unalias, unset, wait -
       bash built-in commands, see bash(1)

BASH BUILTIN COMMANDS
       Unless otherwise noted, each builtin command documented in this section
       as accepting options preceded by - accepts -- to signify the end of the
```

```
mohammadtarek@localhost:~
Mohammadgunzip filetocompress.gz
Mohammad
```

```
Mohammadzip archive.zip filetocompress
  adding: filetocompress (deflated 73%)
Mohammadunzip archive.zip
Archive:  archive.zip
replace filetocompress? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
  inflating: filetocompress
Mohammad
```

2) **What is the command used to view the content of a compressed file.**
   - **zcat**

3) **Backup /etc directory using tar utility.**

```
Mohammadsudo tar -cvf /etctar.tar /etc
[sudo] password for mohammadtarek:
Sorry, try again.
[sudo] password for mohammadtarek:
tar: Removing leading `/' from member names
/etc/
/etc/mtab
/etc/fstab
/etc/crypttab
/etc/lvm/
/etc/lvm/devices/
/etc/lvm/devices/system.devices
/etc/lvm/archive/
/etc/lvm/archive/cs_00000-527924620.vg
/etc/lvm/backup/
/etc/lvm/backup/cs
/etc/lvm/cache/
/etc/lvm/profile/
/etc/lvm/profile/cache-mq.profile
/etc/lvm/profile/cache-smq.profile
/etc/lvm/profile/command_profile_template.profile
```

4) **Starting from your home directory, find all files that were modified in the last two day.**

```
Mohammadpwd
/home/mohammadtarek
Mohammadfind -mtime -2 -type f
./.local/share/gnome-shell/application_state
./.local/share/evolution/addressbook/system/contacts.db
./.local/share/nautilus/tags/meta.db-wal
./.local/share/nautilus/tags/meta.db-shm
./.local/share/nautilus/tags/.meta.isrunning
./.local/share/gvfs-metadata/root
./.local/share/gvfs-metadata/home
./.local/share/gvfs-metadata/root-b37ee7d5.log
./.local/share/gvfs-metadata/home-7d0abd71.log
./.local/share/recently-used.xbel
./.local/state/wireplumber/restore-stream
./.local/state/wireplumber/default-routes
./.config/ibus/bus/10dbb6c979e543b081ba455e779c1ca1-unix-wayland-0
./.config/dconf/user
./.config/gtk-3.0/bookmarks
./.config/enchant/en_US.dic
./.config/enchant/en_US.exc
./.config/gedit/accels
```

5) **Starting from /etc, find files owned by root user.**

```
Mohammadpwd
/etc
Mohammadfind -user root -type f
./fstab
./crypttab
find: ‘./lvm/devices’: Permission denied
find: ‘./lvm/archive’: Permission denied
find: ‘./lvm/backup’: Permission denied
find: ‘./lvm/cache’: Permission denied
./lvm/profile/cache-mq.profile
./lvm/profile/cache-smq.profile
./lvm/profile/command_profile_template.profile
./lvm/profile/lvmdbusd.profile
./lvm/profile/metadata_profile_template.profile
./lvm/profile/thin-generic.profile
./lvm/profile/thin-performance.profile
./lvm/profile/vdo-small.profile
./lvm/lvm.conf
./lvm/lvmlocal.conf
./nvme/hostnqn
./nvme/hostid
./nvme/discovery.conf
./resolv.conf
./dnf/plugins/kpatch.conf
```

6) **Find all directories in your home directory.**

```
Mohammadcd ~
Mohammadpwd
/home/mohammadtarek
Mohammadfind -type d
.
./.mozilla
./.mozilla/extensions
./.mozilla/plugins
./.local
./.local/share
./.local/share/keyrings
./.local/share/gnome-shell
./.local/share/evolution
./.local/share/evolution/addressbook
./.local/share/evolution/addressbook/trash
./.local/share/evolution/addressbook/system
./.local/share/evolution/addressbook/system/photos
./.local/share/evolution/calendar
./.local/share/evolution/calendar/trash
./.local/share/evolution/calendar/system
./.local/share/evolution/mail
./.local/share/evolution/mail/trash
./.local/share/evolution/memos
```

7) **Write a command to search for all files on the system that, its name is ".profile".**

```
Mohammadsudo find -name .profile -type f
Mohammad
```

8) **Identify the file types of the following: /etc/passwd, /dev/pts/0, /etc, /dev/sda**

```
mohammadtarek
Mohammadfile /etc/passwd
/etc/passwd: ASCII text
Mohammadfile /dev/pts/0
/dev/pts/0: character special (136/0)
Mohammadfile /etc
/etc: directory
Mohammadfile /dev/sda
/dev/sda: block special (8/0)
Mohammad
```

9) **List the inode numbers of /, /etc, /etc/hosts.**

```
Mohammad
Mohammadls -i /
  401119 afs      954434 etctar.tar  34093652 mnt      401127 sbin  50331778 var
  401120 bin      401124 home        50628080 opt      401128 srv
     128 boot     401125 lib                1 proc          1 sys
       1 dev      401126 lib64       33685638 root   16777345 tmp
33685633 etc    16973322 media              1 run    16777354 usr
Mohammadls -i /etc
35266246 accountsservice          34401570 magic
34962771 adjtime                  37185950 mailcap
34093622 aliases                  35634687 makedumpfile.conf.sample
36546489 alsa                     36571580 man_db.conf
50655892 alternatives              1581872 mcelog
35299380 anacrontab               36546819 microcode_ctl
```

```
Mohammadls -i /etc/hosts
34093633 /etc/hosts
Mohammad
```

10) **Copy /etc/passwd to your home directory, use the commands diff and cmp, and Edit in the file you copied, and then use these commands again, and check the output.**

```
Mohammaddiff /etc/passwd ~/passwd
Mohammadcmp /etc/passwd ~/passwd
Mohammad
```

```
mohammadtarek@localhost:~
Mohammadvi ~/passwd
Mohammadvi ~/passwd
Mohammaddiff /etc/passwd ~/passwd
35a36
> bro, this is the updated line, idk what will be happened
Mohammadcmp /etc/passwd ~/passwd
cmp: EOF on /etc/passwd after byte 1970, line 35
Mohammad
```

11) **Create a symbolic link of /etc/passwd in /boot.**

12) **Create a hard link of /etc/passwd in /boot. Could you? Why?**

```
Mohammadsudo ln -s  /etc/passwd /boot/fileforslink
[sudo] password for mohammadtarek:
Sorry, try again.
[sudo] password for mohammadtarek:
Mohammadsudo ln  /etc/passwd /boot/fileforslink
ln: failed to create hard link '/boot/fileforslink': File exists
Mohammad
```